3-2 Milestone Two: Enhancement One: Software Design and Engineering

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For this enhancement, I selected my Event Tracking App artifact, which was originally created earlier in the computer science program as part of an Android mobile application project. The purpose of the application is to allow users to create, store, and manage scheduled events using a local SQLite database. The original version of the application included user login functionality, event creation, event display, and event deletion. While the application was functional, it was missing several features that would make it more practical in a real-world setting, including the ability to edit events after they were created, stronger input validation, and automated event reminders.

I selected this artifact for my ePortfolio because it demonstrates several important software development skills that I have built throughout the program, including object-oriented programming, database management, mobile application development, user interface design, and problem solving through software enhancement. This artifact also provided a strong foundation for improvement because the original structure was functional but had room for meaningful expansion. The enhancements I made showcase my ability to improve an existing codebase instead of only creating something from scratch, which is a valuable skill in professional software development. Specifically, I added event editing functionality that allows users to update saved event information, improved input validation to ensure date and time formatting is entered correctly, and implemented local reminder notifications that automatically alert users before an event begins. These changes made the application more user-friendly and aligned it more closely with what would be expected in a production-level scheduling tool.

The enhancements completed for this artifact allowed me to meet the course outcomes I originally planned to focus on in Module One. The strongest outcome demonstrated was designing and evaluating computing solutions using algorithmic principles and software development practices. I improved the application's functionality by designing logic that validates input before data is saved, updates database records when users edit existing events, and schedules notifications based on event timing. This project also supported the outcome related to implementing computer solutions that deliver value through professional software engineering practices. The updates improved both reliability and usability while preserving the original application structure. At this point, I do not have major changes to my original outcome coverage plan because the enhancement work aligned well with what I initially proposed.

Enhancing this artifact gave me a better understanding of what it means to work with and improve an existing software system rather than simply building one from the beginning. One of the biggest lessons I learned was how important planning and code review are when modifying an established codebase. Before making changes, I had to carefully evaluate how the existing files interacted so that updates would not break existing functionality. I also learned more about integrating Android system services, particularly scheduling local notifications and handling application events across multiple components. One challenge I faced was debugging issues caused by placing activity logic inside database classes, which reinforced the importance of keeping responsibilities separated across classes and following clean software design principles. Working through these issues strengthened my debugging skills and helped me better understand how application architecture supports maintainability and scalability. Overall, this enhancement process gave me practical experience with extending software functionality while applying professional development practices that I will carry forward into future projects.